JAMA Psychiatry | Original Investigation

Changes in Genetic Contributions to ASD and ADHD by Year of Diagnosis

Sonja LaBianca, MD, PhD; Mette Lise Lousdal, PhD; Morten Dybdahl Krebs, MD, PhD; Ole Sahlholdt Hansen, MSc; Kajsa-Lotta Georgii Hellberg, BSc; Mischa Lundberg, PhD; Johanne Østerby Sørensen, MD; Jesper R. Gådin, PhD; Henrik Ohlsson, PhD; Jakob Grove, PhD; Anders Børglum, PhD; Esben Agerbo, PhD; Thomas Werge, PhD; Clara Albiñana, PhD; Bjarni J. Vilhjálmsson, PhD; Kenneth S. Kendler, MD; Oleguer Plana-Ripoll, PhD; Andrew J. Schork, PhD; for the iPSYCH Study Consortium and Autism Spectrum Disorder Working Group of the Psychiatric Genomics Consortium

IMPORTANCE The incidences of attention-deficit/hyperactivity disorder (ADHD) and autism spectrum disorder (ASD) have increased markedly over recent decades, raising concerns about the emergence of new risk factors. Current literature typically attributes increased rates to changes in diagnostic practice, stigmatization, and awareness, but critically few studies have explored changes in underlying risk factors.

OBJECTIVE To assess changes in the genetic risk profile of individuals diagnosed with ASD or ADHD by year of incident diagnosis and explore results in comparison with those expected under simulated scenarios.

DESIGN, SETTING, AND PARTICIPANTS This cohort study used data from the Lundbeck Foundation Initiative for Integrative Psychiatric Research (iPSYCH2015) study, a population-based case-cohort in Denmark among individuals with incident diagnoses for ASD and ADHD made from 1994 to 2016. Data were analyzed from January 2024 until September 2025.

EXPOSURE Year of incident diagnosis. Regression models tested changes in the mean genetic risk profile of individuals diagnosed in each consecutive year (1994-2016), adjusting for age, sex, and ancestry.

MAIN OUTCOMES Polygenic scores for psychiatric (ADHD, ASD, depression, bipolar, and schizophrenia) and cognitive-behavioral (addiction, educational attainment, IQ, neuroticism, and risk-taking) outcomes were used to capture genetic risk profiles of diagnosed individuals. Empirical trends were compared with those expected under different simulated drivers of increasing rates.

RESULTS A total of 17 071 individuals with incident diagnoses for ASD (4196 female [24.6%]; mean [SD] age at diagnosis, 9.7 [4.6] years) and 20 111 with diagnoses for ADHD (5960 female [29.6%]; mean [SD] age at diagnosis, 11.0 ([4.7] years) were analyzed. A more recent ADHD diagnosis was associated with decreased genetic risk for ADHD as shown by difference in mean polygenic score (β estimate per 10-year increase, −0.06 SDs; 95% CI, −0.09 to −0.03 SDs; P = .001) and other disorders, including ASD, bipolar, and schizophrenia (eg, ASD: β per 10-year increase, −0.07 SDs; 95% CI, −0.10 to −0.03 SDs; P < .001). Similarly, a more recent ASD diagnosis was associated with decreased genetic risk for ASD (β per 10-year increase, −0.07 SDs; 95% CI, −0.10 to −0.04 SDs; P < .001) and other disorders and traits, including bipolar disorder, schizophrenia, and educational attainment (eg, bipolar disorder: β per 10-year increase, −0.05 SDs; 95% CI, −0.08 to −0.02 SDs; P = .001).

CONCLUSIONS AND RELEVANCE By analyzing multiple polygenic scores together with simulated expectations, this study could disambiguate among competing hypotheses about scenarios that may be associated with increased rates of ADHD and ASD diagnoses. Findings support broadening diagnostic criteria as an explanation for increasing rates, with implications for understanding changes in risk factors and clinical practice.

JAMA Psychiatry. 2026;83(9):921-929. doi:10.1001/jamapsychiatry.2026.1450
Published online June 10, 2026.

Supplemental content

Author Affiliations: Author affiliations are listed at the end of this article.

Group Information: A complete list of the members of the iPSYCH Study Consortium and Autism Spectrum Disorder Working Group of the Psychiatric Genomics Consortium appears in Supplement 3.

Corresponding Authors: Sonja LaBianca, MD, PhD, Institute of Biological Psychiatry, Mental Health Services, Copenhagen University Hospital, Capital Region of Denmark, Amager and Hvidovre Hospital, Mental Health Center Sct. Hans, Institute of Biological Psychiatry, Bldg 134, Boserupvej 2, 4000 Roskilde, Denmark (sonja.labianca@regionh.dk); Andrew J. Schork, PhD, Institute of Biological Psychiatry, Mental Health Services, Copenhagen University Hospital, Capital Region of Denmark, Amager and Hvidovre Hospital, Mental Health Center Sct. Hans, Institute of Biological Psychiatry, Bldg 134, Boserupvej 2, 4000 Roskilde, Denmark (Andrew.Joseph.Schork@regionh.dk).

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

Over recent decades, rates of attention-deficit/hyperactivity disorder (ADHD) and autism spectrum disorder (ASD) diagnoses have increased remarkably, with reports of 4- to 10-fold increases across multiple populations.[1-4] This has sparked debates regarding the emergence of novel environmental risk factors and contributions of changing clinical practice or societal awareness.[5] The nosology of ASD and ADHD has evolved significantly over the last 40 years.[6-8] Initially conceived as distinct, narrowly defined childhood disorders, they are now considered lifelong conditions with a spectrum of manifestations and high rates of co-occurring mental health difficulties.[9,10] Additionally, there has been increased clinical and public awareness, driven in part by concerns of underdiagnosis.[9,11] Undiagnosed ADHD, for example, is associated with negative outcomes, such as unemployment, criminality, and premature mortality.[11,12] Conversely, sharp increases in diagnostic rates have brought about concerns of overdiagnosis,[13-17] which can lead to unnecessary treatments and adverse effects of interventions[18] or can be a consequence of social services requiring a diagnosis to allocate support.[15,17] Furthermore, changes in the environment have been proposed in terms of novel exposures (eg, perinatal acetaminophen exposure[19] or screen time[20]) and societal changes (eg, school environment[21]). Finally, the increased emphasis on diagnosis in females, adolescents, and adults,[2,3] where clinical manifestations differ from previously predominant childhood, male presentations,[10,22-24] further complicates discussions. Although increasing diagnostic rates are often attributed to the interplay of these changes,[9,11] few studies have evaluated whether underlying risk factors are also changing.

ADHD and ASD are highly heritable disorders (heritability estimates of approximately 0.8),[25,26] indicating a large etiological contribution from genes. Environmental exposures, clinical guidelines, and societal factors evolve continuously or may change abruptly. However, an individual's inherited genetic risk remains fixed throughout their lifespan, and population levels are constant over recent decades.[27] This creates a unique opportunity to study the association between time-varying diagnostic trends and time-invariant genetic factors to explore whether diagnoses have evolved to capture different constellations of risk. To our knowledge, only 2 studies, both among twins, have examined this association for ASD[28] and ADHD[29] using indirect measures of genetics and could not detect evidence of changing genetic or environmental risk contributions. Our 2025 study[27] was the first to use polygenic score (PRS), an emerging tool for directly indexing the inherited genetic risk carried by an individual that offers promise in clinical and etiological studies.[30] We showed that individuals diagnosed with any of multiple psychiatric disorders, including ASD and ADHD, carried a lower PRS if born and diagnosed more recently.[27] However, PRSs in that study were restricted to those of the index disorder, and the study did not explore how the association with PRSs for other disorders or traits changed over time. Taken together, the limited studies and current relevance of changing rates of ADHD and ASD necessitate deeper, targeted investigations, and emerging studies[31] suggest that more powerful applications of PRS (ie, using multiple scores to disambiguate among explanations) could add clarity.

Key Points

Question Has the genetic risk profile of individuals diagnosed with autism spectrum disorder (ASD) and attention-deficit/hyperactivity disorder (ADHD) changed as diagnostic rates have increased?

Findings In this cohort study among 17 071 individuals with incident diagnosis of ASD and 20 111 individuals with incident diagnosis of ADHD, genetic contributions to ASD and ADHD diagnoses weakened over the past 2 decades.

Meaning These results suggest that recent increases in ASD and ADHD diagnoses coincided with a broadening of diagnostic criteria.

In this study, we took a novel approach, using multiple PRSs to examine and disambiguate among potential explanations for increasing diagnostic rates: (1) a lowering threshold, consistent with less severe or more heterogeneous symptomatology or emerging environmental risk factors; (2) shifting diagnostic boundaries allowing for different clinical manifestations previously assigned to other diagnoses; or (3) better detection, which implicates catching up on previously undiagnosed individuals. We use the Lundbeck Foundation Initiative for Integrative Psychiatric Research (iPSYCH2015) case-cohort study,[32] the largest genetic research cohort of ADHD and ASD, with diagnoses spanning more than 2 decades.

We analyzed multiple PRSs for psychiatric, cognitive, and behavioral traits (genetic risk profile) to test how each was associated with year of diagnosis. In simulations, we addressed temporal changes to the genetic risk profile under potential explanations for increased rates and used this framework to interpret our results.

Methods

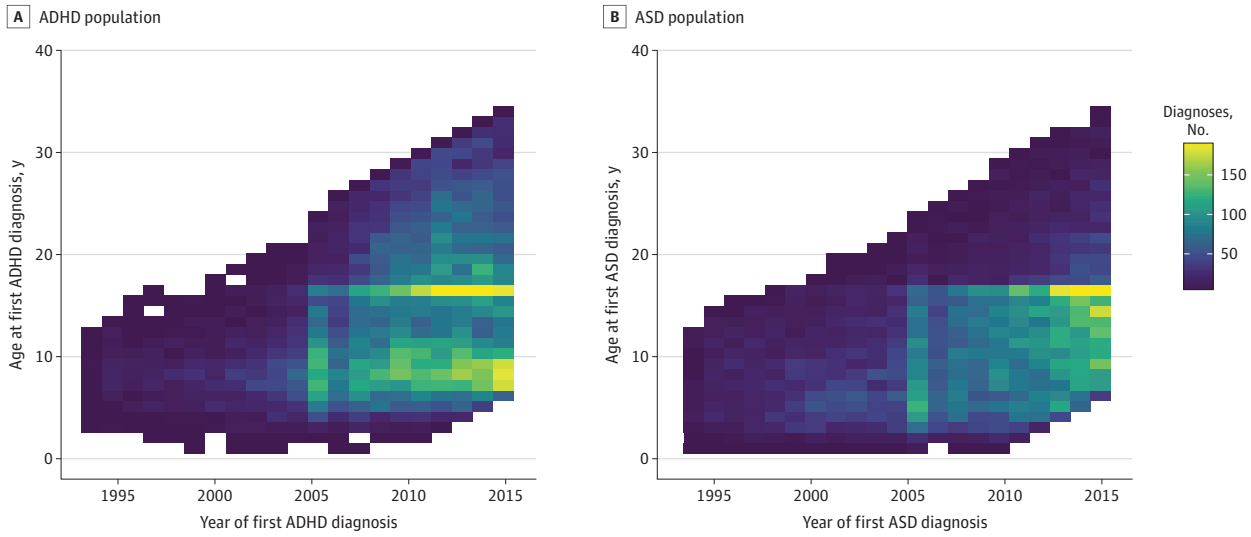
Study Population

This cohort study used data from iPSYCH2015,[32,33] a Danish case-cohort study using national biobanks and registers. The source population consists of all 1 657 449 singleton births May 1, 1981, to December 31, 2008, with the individual alive and residing in Denmark at age 1 year. The sample of 143 265 individuals selected all 93 608 incident diagnoses of major psychiatric disorders through 2015 and a random population cohort of 50 615 individuals. Over the entire study period, only the *International Statistical Classification of Diseases and Related Health Problems, Tenth Revision* (*ICD-10*) was implemented in Denmark. We included 29 668 individuals with a registered diagnosis of ADHD (*ICD-10* code F90.0) and 24 975 individuals with ASD (*ICD-10* codes F84.0, F84.1, F84.5, F84.8, or F84.9) in the Danish Psychiatric Central Research Register[34] between January 1, 1994, and December 31, 2015 (individuals could have both diagnoses). All personal information from registers is anonymized when used for research purposes, and the project was approved by the Danish Data Protection Agency; hence, according to Danish legislation, informed consent from

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

Figure 1. Heat Map of Study Population Distribution of Year and Age



Each cell represents counts of individuals diagnosed in that particular year and age, with cells restricted to those with more than 4 individuals due to data protection. ADHD indicates attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder.

participants was not required. Use of iPSYCH2015 data follows Danish Scientific Ethics Committee, Danish Health Data Authority, Danish Data Protection Agency, and Danish Neonatal Screening Biobank Steering Committee guidelines. Data access was via secure portals in accordance with guidelines set by the Danish Data Protection Agency, Danish Health Data Authority, and Statistics Denmark. This study followed the Strengthening the Reporting of Observational Studies in Epidemiology (STROBE) reporting guideline. Diagnostic rates for ADHD and ASD were displayed in 2-dimensional histograms (**Figure 1**) using the R statistical software package ggplot2, presenting 1 × 1-year bins with year of first diagnosis along the x-axis and age at first diagnosis along the y-axis.

Genotyping and Quality Controls

DNA was extracted and amplified from dried blood spots at the Danish Neonatal Screening Biobank[35] and genotyped using the Illumina Global Screening or the Infinium PsychChip v1.0 Array. Data quality control is described elsewhere.[32,33] Imputation followed the Rapid Imputation for Consortias Pipeline (RICOPILI) pipeline[36] using the Haplotype Reference Consortium panel.[37] Ancestry principal component analysis was conducted to create the largest homogenous group for analysis, a sample of broadly northern European genetic ancestry, and retain 10 principal components as analytical covariates.

Genetic Risk Profiles

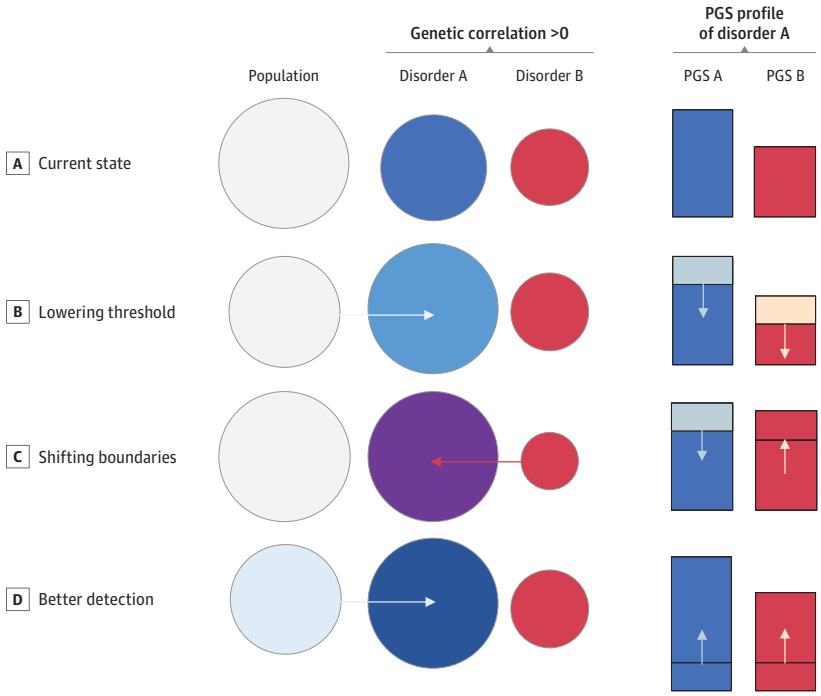
Genetic risk profiles included 10 PRSs constructed using results from genome-wide association studies (GWAS) in public repositories. GWAS included 5 psychiatric disorders (ADHD,[38] ASD,[39] bipolar disorder,[40] major depressive disorder,[41] and schizophrenia[42]) and 5 cognitive or behavioral traits (educational attainment,[43] IQ,[44] risk-taking,[45] neuroticism,[46] and addiction[47]) (eTable 1 in Supplement 1). For psychiatric disorders, we used the meta-PRS method[48] to construct PRSs that were weighted combinations of PRSs trained using published GWAS independent of iPSYCH (*external PRS*;eTable 1 in Supplement 1) and PRSs trained by cross-validation within iPSYCH (*internal PRS*; eTable 1 in Supplement 1). Briefly, for each iteration of cross-validation, single-nucleotide variation weights were estimated by GWAS with BOLT-LMM in 80% of iPSYCH and used to train a PRS in the held-out 20%, with sample size weights used to combine internal and external PRSs. For cognitive and behavioral traits, where phenotypes were not available in iPSYCH, PRSs were computed with LDpred2[49] based on external GWAS only (eTable 1 in Supplement 1).

Statistical Analysis

Linear regressions were fitted with each PRS separately, scaled to the population cohort as the outcome and year of first diagnosis, age, sex as recorded in registers, and 10 principal components as covariates. We adjusted for age to account for increasing age at diagnosis in more recent years.[3] Year trends for PRS were depicted by projecting expected PRS from the fitted model over a range of years, holding other covariables constant at their mean. Post hoc sensitivity analyses explored the robustness of results. First, because age and year are confounded in a birth cohort,[50] goodness of fit and consistency of effect sizes for year adjusted by age and other covariates were compared vs alternative models, including covariates only; covariates and year; covariates and age; covariates, age, and birth cohort; or covariates and birth cohort. In addition, we display our data in descriptive bar plots for which we stratified our patient population into 6 subgroups representing children (aged <14 years at diagnosis), adolescents (aged 14-17 years at diagnosis), and adults (aged >17 years at diagnosis) diagnosed in 3 periods (1995-2001, 2002-2008, and 2009-2015). We scaled

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

Figure 2. Simulation Graphs of Temporal Changes to Genetic Risk Profiles by Explanation of Increasing Diagnostic Rates



Genetic risk profiles are expected to have different associations under different scenarios of increasing diagnoses. A profile of 2 polygenic scores (PRSs) is considered, 1 for an index disorder (A, blue) and 1 for a second disorder (B, red) that is assumed to share genetic contributions with the index disorder (genetic correlation, r_G, >0). The expected profile for a current case with disorder A has high levels of PRS for disorder A and modest levels for disorder B due to the r_G (A). If the number with disorder A is expanded by a lowering threshold, both PRS A and PRS B are expected to decrease (B). If expansion is due to shifting boundaries, PRS A is expected to decrease, while PRS B would increase. If expansion is due to better detection, PRS A and PRS B will appear to increase because controls are expected to decrease (D).

all 10 PRSs to the mean level in the subgroup of children diagnosed between 1995 and 2001 and tested for significant changes across remaining subgroups in analysis of variance. Second, nonlinear associations of year were explored by adding quadratic terms and comparing the fit with linear-only models. Sex differences were tested by including interaction terms (sex by year) and comparing the model fit. Third, associations of year with meta-PRS[48] were compared vs those with PRS computed via LDpred2[49] using the same GWAS excluding internal iPSYCH data (eTable 1 in Supplement 1). Fourth, a set of alternative PRSs were computed with LDpred2 using GWAS from public repositories excluding iPSYCH (eTable 1 in Supplement 1 and eNote 1 in Supplement 2). We tested the association of each alternative PRS with year of diagnosis in ADHD or ASD, respectively, and compared them with our main results. Finally, we show the distribution of *ICD-10* diagnostic subcodes in ASD over time to explore changes that could impact our results. For all analysis, the significance was determined by Bonferroni adjustment for 2 disorders and 10 PRSs (2-sided $P < .05/[2 \times 10] = .0025$). All analyses were conducted using R statistical software version 4.1.3 (R Project for Statistical Computing). Data were analyzed from January 2024 until September 2025.

Results

Diagnostic Rates of ADHD and ASD in Denmark, 1994-2015

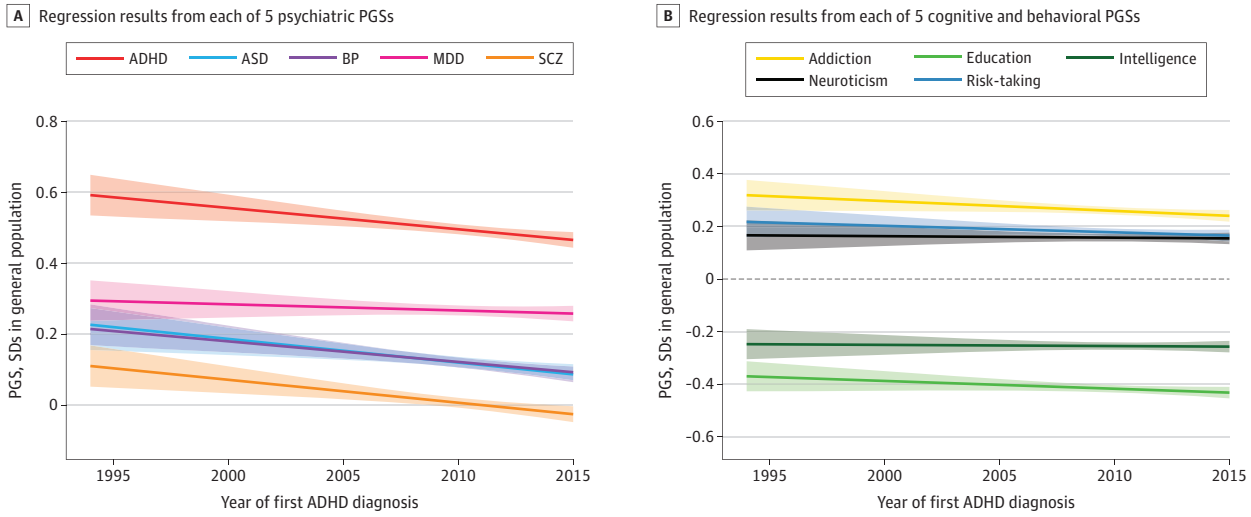
Previous reports have shown an increasing diagnostic rate of ASD and ADHD.[3] We depict these trends in our specific analysis samples. From the iPSYCH case-cohort study, we included 20 111 individuals (5960 female [29.6%]; mean [SD] age at diagnosis, 11.0 ([4.7] years) first diagnosed with ADHD between January 1, 1994, and December 31, 2015 (Figure 1A; eTable 2 in Supplement 1) and 17 071 individuals (4196 female [24.6%]; mean [SD] age at diagnosis, 9.7 [4.6] years) first diagnosed with ASD in the same period (Figure 1B; eTable 3 in Supplement 1). For both disorders, there was an appreciable increase in the number of diagnoses at a given age with increasing year, as seen in the increasing brightness from left to right in Figure 1.

Simulation of Estimated Changes to Genetic Risk Profiles

We conducted simulations to explore trends in genetic risk profiles that were estimated to coincide with increases in diagnostic rates associated with different hypothetical scenarios (**Figure 2**; eNote 2 in Supplement 2). Our model posits that new cases were diagnosed from 1 of 3 undiagnosed segments of populations: (1) individuals from the general population with milder forms of the index disorder (referred to as *lowering threshold*), (2) individuals with a set of symptoms consistent with or previously referred to as a second disorder (referred to as *shifting boundaries*), and (3) individuals with a set of symptoms consistent with the index disorder but who were not previously diagnosed (referred to as *better detection*). Under a lowering threshold scenario, the index PRS was reduced. The secondary-disorder PRS changes according to the genetic correlation between disorders; in this scenario, it will be reduced due to positive genetic correlation. For shifting boundaries, we expect the index disorder PRS to similarly decrease but PRSs for secondary disorders to increase, regard-

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

Figure 3. Line Graphs of Regression Trends for Genetic Risk Profiles in Attention-Deficit/Hyperactivity Disorder (ADHD) Over Time



Genetic risk profiles in 20 111 patients with ADHD reveal decreasing trends for year of diagnosis between 1994 and 2015. The polygenic score (PRS) level is shown in SDs to the general population. Each line represents regression results from each of 5 psychiatric PRSs (A) and 5 cognitive or behavioral PRSs (B) estimated by year while adjusting for age and covariate shown in the **Table**. ASD indicates autism spectrum disorder; BP, bipolar disorder, MDD, major depressive disorder; SCZ, schizophrenia spectrum disorder.

Table. Linear Regression Estimating PRS From Year of Diagnosis

Disorder or trait	PRS, β per 10-y increase in y of ADHD diagnosis, SD[a]	*P* value	PRS, β per 10-y increase in y of ASD diagnosis, SD[a]	*P* value[b]
ADHD	−0.06 (−0.09 to −0.03)	.001	0.04 (0.01 to 0.07)	.005
ASD	−0.07 (−0.10 to −0.03)	.001	−0.07 (−0.10 to −0.04)	.001
BP	−0.06 (−0.09 to −0.02)	.001	−0.05 (−0.08 to −0.02)	.001
MDD	−0.02 (−0.05 to 0.01)	.32	0.01 (−0.02 to 0.04)	.66
SCZ	−0.06 (−0.10 to −0.03)	.001	−0.05 (−0.08 to −0.02)	.001
Addiction	−0.04 (−0.07 to −0.00)	.04	−0.02 (−0.05 to 0.01)	.26
Education	−0.03 (−0.06 to 0.00)	.09	−0.10 (−0.13 to −0.07)	.001
Intelligence	−0.00 (−0.04 to 0.03)	.80	−0.03 (−0.07 to −0.00)	.03
Risk-taking	−0.02 (−0.06 to 0.01)	.17	−0.04 (−0.07 to −0.00)	.03
Neuroticism	−0.01 (−0.04 to 0.03)	.75	0.01 (−0.02 to 0.04)	.38

Abbreviations: ADHD, attention-deficit/hyperactivity disorder; ASD, autism spectrum disorder; BP, bipolar disorder; MDD, major depressive disorder; PRS, polygenic score; SCZ, schizophrenia spectrum disorder.

[a] Covariates are sex and 10 principal components of ancestry. Estimated mean PRS change is given in SDs per 10 years.

[b] The significance level after Bonferroni adjustment was .0025.

less of genetic correlation. For better detection, we expect increases in differences in PRS between cases and controls, with an increase of secondary scores in proportion to genetic correlation. These vignettes were confirmed with simulation experiments (eNote 2 in Supplement 1).

ADHD Diagnoses and Genetic Risk by Year

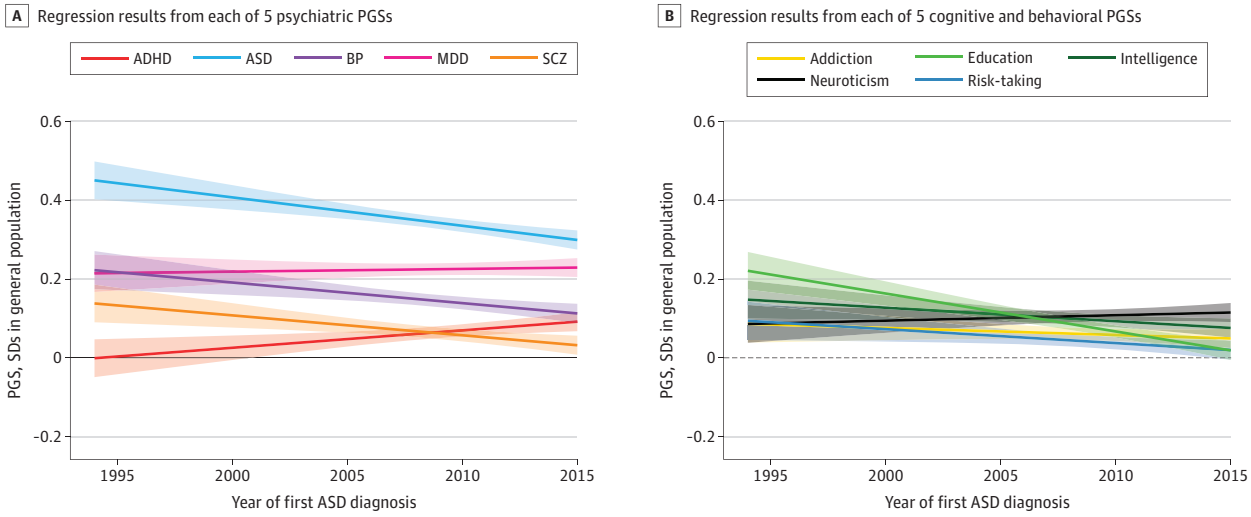
Regression models (**Figure 3**, **Table**; eTables 4-13 in Supplement 1) suggested that the mean PRS of ADHD cases decreased significantly with increasing year of diagnosis (β per 10-year increase, −0.06 SDs in the general population; 95% CI, −0.09 to −0.03 SDs; *P* = .001) but not with increasing age (β estimate per 10-year increase in age at first diagnosis, −0.00 SDs; 95% CI, −0.02 to 0.02 SDs; *P* = .81). The mean level of other PRSs also decreased with increasing year of diagnosis, including ASD (β per 10-year increase, −0.07 SDs; 95% CI, −0.10 to −0.03 SDs; *P* < .001), bipolar disorder (β per 10-year increase, −0.06 SDs; 95% CI, −0.09 to −0.02 SDs; *P* = .001) and schizophrenia (β per 10-year increase, −0.06; 95% CI, −0.10 to −0.03 SDs; *P* < .001). The directions of age associations were inconsistent across PRS; while major depressive disorder, bipolar disorder, schizophrenia, addiction, and neuroticism significantly increased, ASD, educational attainment, and IQ decreased (eFigure 1 in Supplement 2 and eTables 4-13 in Supplement 1). These results appeared consistent with a lowering threshold for increasing year of ADHD diagnosis (Figure 2 and Figure 3) and reflected unmodeled trends in underlying data (eFigure 2 in Supplement 2).

ASD Diagnoses and Genetic Risk by Year

Regression models in ASD (**Figure 4** and Table; eTables 14-23 in Supplement 1) also suggested that ASD PRS significantly decreased with increasing year of diagnosis (β estimate per 10-year increase, −0.07 SDs in the general population; 95% CI, −0.10 to −0.04 SDs; *P* < .001) but not age (β per 10-year increase in age at first diagnosis, −0.01 SDs; 95% CI, −0.04 to 0.02 SDs; *P* = .460). With increasing year of diagnosis, we also observed significant decreases in other PRSs, including bipolar

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

Figure 4. Line Graphs of Regression Trends for Genetic Risk Profiles in Autism Spectrum Disorder (ASD) Over Time



Genetic risk profiles in 17 071 patients with ASD reveal decreasing trends for year of diagnosis between 1994 and 2015. The polygenic score (PRS) level is shown in SDs to the general population. Each line represents regression results from each of 5 psychiatric PRSs (A) and 5 cognitive or behavioral PRSs (B) estimated by year while adjusting for age and covariate shown in the **Table**. ASD indicates autism spectrum disorder; BP, bipolar disorder, MDD, major depressive disorder; SCZ, schizophrenia spectrum disorder.

disorder (β per 10-year increase, −0.05 SDs; 95% CI, −0.08 to −0.02 SDs; P = .001), schizophrenia (β per 10-year increase, −0.05 SDs; 95% CI, −0.08 to −0.02 SDs; P = .001), and educational attainment (defined in discovery GWAS as a continuous variable presenting years of schooling; β, −0.10 SDs; 95% CI, −0.13 to −0.07 SDs; P < .001). We observed a significant increase in ADHD PRS before adjustment for multiple testing (β per 10-year increase, 0.04 SDs; 95% CI, 0.01 to 0.07 SDs; P = .005), but this was not significant after adjustment. The directions of age associations in patients with ASD were inconsistent across PRSs; while major depressive disorder, addiction, and neuroticism significantly increased, educational attainment decreased (eFigure 3 in Supplement 2 and eTables 14-23 in Supplement 1). For ASD, our results also appeared consistent with a lowering threshold with increasing year (Figure 2 and Figure 4) and reflected trends in underlying data (eFigure 4 in Supplement 2).

Sensitivity Analysis

To test the robustness of our results, we conducted several sensitivity analyses (eTables 24-32 in Supplement 1 and eFigures 1-10 in Supplement 2). Most critically, results were robust to training GWAS phenotype definition in 2 ways. They were consistent when PRSs were trained on only external summary statistics (ie, not using meta-PRS to incorporate information from iPSYCH in the PRS) (eTable 24 in Supplement 1 and eFigure 6 in Supplement 2 for ADHD and eTable 28 in Supplement 1 and eFigure 8 in Supplement 2 for ASD). They were also consistent when using an alternate set of external GWAS (eTable 25 in Supplement 1 and eFigure 5 in Supplement 2 for ADHD and eTable 29 in Supplement 1 and eFigure 7 in Supplement 2 for ASD). One exception was an attenuation of the effect size for year of ADHD diagnosis and ADHD PRS when limiting PRS training to external GWAS (eTable 24 in Supplement 1 and eFigure 6 in Supplement 2 for the original PRS and eTable 25 in Supplement 1 and eFigure 5 in Supplement 2 for the alternative PRS), noting that this also reduced the training sample size (eTable 1 in Supplement 1). Statistical adjustments to account for the correlation between age and year (Figure 1) produced trends that were stable or, in a few cases, better representative of underlying data (eFigures 2 and 5 and eFigures 4 and 7 in Supplement 2 for ADHD and ASD, respectively). Incorporating nonlinear terms for year (eTables 26 and 30 in Supplement 1) or interactions with sex (eTables 27 and 31 in Supplement 1) had minimal impact on outcomes. Finally, *ICD-10* diagnostic subcode distribution in ASD remained stable (eTable 32 in Supplement 1 and eFigures 9-10 in Supplement 2).

Discussion

In this cohort study, we found a decreasing genetic risk by year of diagnosis of ASD and ADHD over a period in which the diagnostic rate increased more than 4-fold. We interpret these results as consistent with prior studies suggesting a broadening of diagnostic criteria is the most plausible driver of increasing rates of both diagnoses.

Our approach is informative for differentiating among certain higher-level hypotheses, but we are limited in our ability to draw strong conclusions as to why a lowering threshold appears most consistent with our data. For example, the genetic risk profile can be reduced by misdiagnosis or overdiagnosis.[13,15-17] In this study, individuals without ADHD or ASD (and thus with low genetic risk) or milder and more heterogeneous presentations (assuming severity and genetic

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

risk are correlated[38,39]) would lower the mean risk in genetic profiles of cases. An equivalent framing is that a lowering threshold is occurring because previously only individuals with the most extreme genetic liability were diagnosed; this is because although the profile decreased, PRS levels in new cases with a lower mean genetic liability remained significantly higher than observed in the general population. Likewise, an increased burden of environmental causes could be consistent with a lowering threshold given that our study can inform only on the genetic threshold. For example, increased exposure to screens and social media[20] or changes to early education environments[21] have been proposed as risk factors, although evidence is neither consistently reported nor required to explain current trends. It is difficult to disambiguate among these hypotheses retrospectively; however, intervention studies, prospective longitudinal cohorts, and the extraction of detailed measures from, for example, clinical notes could help.

Evolving nosology and changing clinical practice around adult diagnosis and recognition in females is reflected in time trends of the diagnosed population.[3] As such, we adjusted for age in our models and tested for sex interaction. While we did not observe sex differences in the genetic risk profile over time, PRS associations for age revealed a transition with increasing age toward less neurodevelopmental and more internalizing and substance use genetic risk but no changes in the index disorder PRS, consistent with our previous results for ADHD.[32] This preliminary finding suggests that genetic associations with other clinical features change with age, which could be explored in future studies.

This study significantly extends the limited body of work associating genetic risk with diagnostic time trends (ie, year) in ADHD and ASD. Our results draw different conclusions than 2 previous studies using Swedish twin registers, which suggested that genetic and environmental contributions to ADHD[29] and ASD[28] have not changed markedly over time. We believe that direct measures of genetics (eg, PRS) provide a more powerful index, and this difference in design may be critical. This study adds significantly to our previous work,[27] in which we observed a decreased index PRS for ADHD and ASD diagnosed in individuals from more recent birth cohorts. We confirm initial support of lowering threshold with consistent evidence from multiple PRS and reveal unique associations with age.

Our results have important implications for the design, interpretation, and application of large-scale genetic epidemiological and gene-mapping studies. Few studies[27-29] have considered the year of first diagnosis as a contributor to differences among cohorts. Our results in this study and previously reported changes[3] over time in epidemiological or clinical characteristics (eg, prevalence, age, and sex ratio) suggest that time trends could contribute to cross-cohort heterogeneity. This may be especially true when considering age of onset or first diagnosis given that there are well-described time trends across all disorders[3] and given that age (ie, age at diagnosis) can be confounded with period (ie, year of diagnosis) and cohort (ie, birth year) in ways that are difficult to disambiguate.[50] Such differences could impact GWAS, which map different polygenes, if cases of ADHD or ASD were sampled using diagnoses made in different year windows, for example. Speculatively, diagnostic time trends could also have implications for the design and results of studies that involve downstream applications of GWAS (eg, PRS associations with other outcomes, GWAS revealing different patterns of genetic correlations, or different pathways being enriched for polygenic burden). As meta-analyses become broader and more inclusive and aggregate across longer times, understanding how time trends may impact our inferences into etiology and estimative ability is important.

Limitations

Our study has several limitations. First, data from Danish registers do not include diagnoses made in primary care or by psychiatrists in specialized care outside hospital inpatient and outpatient facilities. Second, birth year truncation is embedded in the iPSYCH case-cohort design, which should be considered when generalizing findings. However, we note that our previous work[27] included restrictions to limit this and that results are consistent. Third, the iPSYCH case-cohort is relatively young (ie, few patients older than 35 years), which limits our ability to study a full life-course and earlier birth cohorts. However, the recent introduction of ADHD and ASD and the typically early age at diagnosis for these conditions are reassuring for our design. Fourth, published GWAS will have ascertained cases from earlier periods and therefore may better resemble earlier-diagnosed cases in our cohort. This increased similarity could result in PRSs trained on external GWAS associating more strongly with earlier-diagnosed cases in our study. There was consistency in associations observed when using meta-PRS, which includes cross-validated training GWAS data from iPSYCH, where most cases were diagnosed in recent periods, and when using PRS trained using these external GWAS data, where cases are likely diagnosed from earlier periods. We believe this consistency suggests that our results were not driven by spurious time-dependent ascertainment. Fifth, the association of year of ADHD diagnosis with ADHD PRS was driven by internal (iPSYCH) training samples, which are 55% to 70% of cases in recent ADHD GWAS. Further replication with larger external training samples, when they become available, could contextualize generalizability of our results to other cohorts. Sixth, our analysis was limited to Denmark using individuals of broadly Northern European ancestry. Given the interest in this topic, replication to other populations, countries, societies, racial and ethnic groups, and health care settings is important.

Conclusions

This cohort study offers a new perspective for ongoing debates about increasing diagnostic rates of ASD and ADHD. By analyzing genetic risk profiles across 2 decades of diagnoses in a large population-based case cohort, we demonstrated that individuals diagnosed more recently had systematically lower genetic risk. These findings suggest that clinical practice and diagnostic criteria have changed over time to include a broader phenotypic manifestation of ASD and ADHD.

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.
Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

ARTICLE INFORMATION

Accepted for Publication: March 22, 2026.

Published Online: June 10, 2026. doi:10.1001/jamapsychiatry.2026.1450

Author Affiliations: Institute of Biological Psychiatry, Mental Health Services, Copenhagen University Hospital, Roskilde, Denmark (LaBianca, Dybdahl Krebs, Georgii Hellberg, Lundberg, Østerby Sørensen, Gådin, Werge, Schork); National Centre for Register-Based Research, Department of Public Health, Aarhus University, Aarhus, Denmark (Lousdal, Sahlholdt Hansen, Agerbo, Albiñana, Vilhjálmsson, Plana-Ripoll); Bioinformatics and Bioinformatics PhD Graduate Programme, Graduate School of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark (Georgii Hellberg); Psychiatry PhD Graduate Programme, Graduate School of Health and Medical Sciences, University of Copenhagen, Copenhagen, Denmark (Østerby Sørensen); Center for Primary Health Care Research, Lund University, Malmö, Sweden (Ohlsson); Department of Biomedicine-Human Genetics, Aarhus University, Aarhus, Denmark (Grove, Børglum); Bioinformatics Research Centre, Aarhus University, Aarhus, Denmark (Grove); The Lundbeck Foundation Initiative for Integrative Psychiatric Research, Denmark (Grove, Børglum); Center for Genomics and Personalized Medicine, Aarhus, Denmark (Grove, Børglum); Big Data Institute, University of Oxford, Oxford, United Kingdom (Albiñana, Vilhjálmsson); Bioinformatics Research Centre, Aarhus University, Aarhus, Denmark (Vilhjálmsson); Virginia Institute for Psychiatric and Behavioral Genetics, Virginia Commonwealth University, Richmond (Kendler); Barcelona Institute for Global Health, Barcelona, Spain (Plana-Ripoll); Department of Psychiatry, Virginia Commonwealth University, Richmond (Kendler).

Author Contributions: Dr LaBianca had full access to all of the data in the study and takes responsibility for the integrity of the data and the accuracy of the data analysis. Drs Plana-Ripoll and Schork jointly supervised this work.
Concept and design: LaBianca, Lousdal, Agerbo, Plana-Ripoll, Schork.
Acquisition, analysis, or interpretation of data: All authors.
Drafting of the manuscript: LaBianca, Georgii Hellberg, Schork.
Critical review of the manuscript for important intellectual content: LaBianca, Lousdal, Dybdahl Krebs, Sahlholdt Hansen, Lundberg, Østerby Sørensen, Gådin, Ohlsson, Grove, D. Børglum, Agerbo, Werge, Albiñana, Vilhjálmsson, Kendler, Plana-Ripoll, Schork.
Statistical analysis: LaBianca, Lousdal, Dybdahl Krebs, Sahlholdt Hansen, Lundberg, Ohlsson, Grove, Agerbo, Plana-Ripoll, Schork.
Obtained funding: LaBianca, D. Børglum, Werge, Schork.
Administrative, technical, or material support: Georgii Hellberg, Lundberg, Gådin, Agerbo, Werge, Schork.
Supervision: Grove, Agerbo, Werge, Vilhjálmsson, Kendler, Plana-Ripoll, Schork.

Conflict of Interest Disclosures: Dr D. Børglum reported receiving personal fees from H. Lundbeck outside the submitted work. Dr Vilhjálmsson reported membership in the scientific advisory board for Allelica. No other disclosures were reported.

Funding/Support: The study was specifically supported by grant R4A92 from the Research Fund of the Mental Health Services–Capital Region of Denmark to Dr LaBianca. Dr Dybdahl Krebs received funding from grant R450-2023-1447 from the Lundbeck Foundation. Dr Plana-Ripoll received funding from grants 2066-00009B and 4309-00026B from the Independent Research Fund Denmark. Dr Schork received funding from grant R335-2019-2318 from the Lundbeck Foundation. Drs Kendler and Schork are supported by grant R01MH130581 from the US National Institute of Mental Health (NIMH). Dr Vilhjalmsson is supported by grants R335-2019-2339 from the Lundbeck Foundation and 2034-00241B from the Independent Research Fund Denmark. The iPSYCH team was supported by grants R102-A9118, R155-2014- 1724, and R248-2017-2003 from the Lundbeck Foundation and 1U01MH109514-01 and 1R01MH124851-01 from the US National Institutes of Health NIMH and by the Universities and University Hospitals of Aarhus and Copenhagen. The Danish National Biobank resource was supported by the Novo Nordisk Foundation. High-performance computer capacity for handling and statistical analysis of iPSYCH data on the GenomeDK high-performance computing facility was provided by the Center for Genomics and Personalized Medicine and Centre for Integrative Sequencing, iSEQ, Aarhus University, Denmark.

Role of the Funder/Sponsor: The funders had no role in the design and conduct of the study; collection, management, analysis, and interpretation of the data; preparation, review, or approval of the manuscript; and decision to submit the manuscript for publication.

Group Information: A complete list of the members of the iPSYCH Study Consortium and Autism Spectrum Disorder Working Group of the Psychiatric Genomics Consortium appears in Supplement 3.

Data Sharing Statement: See Supplement 4.

Additional Contributions: The authors thank Nicholas J. Schork, PhD (The Translational Genomics Research Institute, Phoenix, Arizona), for providing feedback and discussions. This individual was not compensated for this work.

REFERENCES

1. Xu G, Strathearn L, Liu B, Yang B, Bao W. Twenty-Year trends in diagnosed attention-deficit/hyperactivity disorder among US Children and adolescents, 1997-2016. *JAMA Netw Open*. 2018;1(4):e181471-e181471. doi:10.1001/jamanetworkopen.2018.1471

2. Russell G, Stapley S, Newlove-Delgado T, et al. Time trends in autism diagnosis over 20 years: a UK population-based cohort study. *J Child Psychol Psychiatry*. 2022;63(6):674-682. doi:10.1111/jcpp.13505

3. Momen NC, Beck C, Lousdal ML, et al. Mental health disorder trends in Denmark according to age, calendar period, and birth cohort. *JAMA Psychiatry*. 2025;82(2):161-170. doi:10.1001/jamapsychiatry.2024.3723

4. Atladottir HO, Gyllenberg D, Langridge A, et al. The increasing prevalence of reported diagnoses of childhood psychiatric disorders: a descriptive multinational comparison. *Eur Child Adolesc Psychiatry*. 2015;24(2):173-183. doi:10.1007/s00787-014-0553-8

5. Pappas S. RFK, Jr., is wrong about cause of rising autism rates, scientists say. *Scientific American*. April 17, 2025. Accessed July 3, 2025. https://www.scientificamerican.com/article/the-real-reason-autism-rates-are-rising/

6. World Health Organization. *The ICD-10 Classification of Mental and Behavioural Disorders*. World Health Organization; 1992. Accessed May 5, 2026. https://cdn.who.int/media/docs/default-source/classification/other-classifications/9241544228_eng.pdf

7. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders*. 4th ed. American Psychiatric Association; 1994.

8. American Psychiatric Association. *Diagnostic and Statistical Manual of Mental Disorders (DSM-5)*. 5th ed. American Psychiatric Association; 2013.

9. Happé F, Frith U. Annual research review: looking back to look forward—changes in the concept of autism and implications for future research. *J Child Psychol Psychiatry*. 2020;61(3):218-232. doi:10.1111/jcpp.13176

10. Franke B, Michelini G, Asherson P, et al. Live fast, die young: a review on the developmental trajectories of ADHD across the lifespan. *Eur Neuropsychopharmacol*. 2018;28(10):1059-1088. doi:10.1016/j.euroneuro.2018.08.001

11. Abdelnour E, Jansen MO, Gold JA. ADHD diagnostic trends: increased recognition or overdiagnosis? *Mo Med*. 2022;119(5):467-473.

12. Daley D, Jacobsen RH, Lange AM, Sørensen A, Walldorf J. *Costing Adult Attention Deficit Disorder: Impact on the Individual and Society*. Oxford University Press; 2015:232. doi:10.1093/acprof:oso/9780198745556.001.0001

13. Kazda L, Bell K, Thomas R, McGeechan K, Sims R, Barratt A. Overdiagnosis of attention-deficit/hyperactivity disorder in children and adolescents: a systematic scoping review. *JAMA Netw Open*. 2021;4(4):e215335. doi:10.1001/jamanetworkopen.2021.5335

14. Fombonne E. Editorial: is autism overdiagnosed? *J Child Psychol Psychiatry*. 2023;64(5):711-714. doi:10.1111/jcpp.13806

15. Arvidsson O, Gillberg C, Lichtenstein P, Lundström S. Secular changes in the symptom level of clinically diagnosed autism. *J Child Psychol Psychiatry*. 2018;59(7):744-751. doi:10.1111/jcpp.12864

16. Kazda L, McGeechan K, Bell K, Thomas R, Barratt A. Increased diagnosis of attention-deficit hyperactivity disorder despite stable hyperactive/inattentive behaviours: evidence from two birth cohorts of Australian children. *J Child Psychol Psychiatry*. 2023;64(8):1140-1148. doi:10.1111/jcpp.13700

17. Rydell M, Lundström S, Gillberg C, Lichtenstein P, Larsson H. Has the attention deficit hyperactivity disorder phenotype become more common in children between 2004 and 2014: trends over 10 years from a Swedish general population sample. *J Child Psychol Psychiatry*. 2018;59(8):863-871. doi:10.1111/jcpp.12882

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026

18. Layton TJ, Barnett ML, Hicks TR, Jena AB. Attention deficit-hyperactivity disorder and month of school enrollment. *N Engl J Med*. 2018;379(22):2122-2130. doi:10.1056/NEJMoa1806828

19. Prada D, Ritz B, Bauer AZ, Baccarelli AA. Evaluation of the evidence on acetaminophen use and neurodevelopmental disorders using the Navigation Guide methodology. *Environ Health*. 2025;24(1):56. doi:10.1186/s12940-025-01208-0

20. Madelaire LF, Jeppesen P, Hansen ASK, et al. Screen use and the mental health of children and young people. Skærmbrug og børn og unges mentale sundhed. *Ugeskr Laeger*. 2024;186(48):V05240342. doi:10.61409/V05240342

21. Cheesman R, Eilertsen EM, Ayorech Z, et al. How interactions between ADHD and schools affect educational achievement: a family-based genetically sensitive study. *J Child Psychol Psychiatry*. 2022;63(10):1174-1185. doi:10.1111/jcpp.13656

22. Jadav N, Bal VH. Associations between co-occurring conditions and age of autism diagnosis: implications for mental health training and adult autism research. *Autism Res*. 2022;15(11):2112-2125. doi:10.1002/aur.2808

23. Lai MC, Szatmari P. Sex and gender impacts on the behavioural presentation and recognition of autism. *Curr Opin Psychiatry*. 2020;33(2):117-123. doi:10.1097/YCO.0000000000000575

24. Williams T, Horstmann L, Kayani L, et al. An item-level systematic review of the presentation of ADHD in females. *Neurosci Biobehav Rev*. 2025;171:106064. doi:10.1016/j.neubiorev.2025.106064

25. Faraone SV, Bellgrove MA, Brikell I, et al. Attention-deficit/hyperactivity disorder. *Nat Rev Dis Primers*. 2024;10(1):11. doi:10.1038/s41572-024-00495-0

26. Ramaswami G, Geschwind DH. Genetics of autism spectrum disorder. In: Geschwind DH, Paulson HL, Klein C, eds. *Handbook of Clinical Neurology*. Vol 147. Elsevier B.V; 2018:321-329.

27. Lousdal ML, LaBianca S, Agerbo E, et al. Changes in polygenic burden for psychiatric disorders across two decades of birth cohorts. *Nat Ment Health*. 2025;3(9):1037-1045. doi:10.1038/s44220-025-00478-4

28. Taylor MJ, Rosenqvist MA, Larsson H, et al. Etiology of autism spectrum disorders and autistic traits over time. *JAMA Psychiatry*. 2020;77(9):936-943. doi:10.1001/jamapsychiatry.2020.0680

29. Taylor MJ, Martin J, Butwicka A, et al. A twin study of genetic and environmental contributions to attention-deficit/hyperactivity disorder over time. *J Child Psychol Psychiatry*. 2023;64(11):1608-1616. doi:10.1111/jcpp.13854

30. Wray NR, Lin T, Austin J, et al. From basic science to clinical application of polygenic risk scores: a primer. *JAMA Psychiatry*. 2021;78(1):101-109. doi:10.1001/jamapsychiatry.2020.3049

31. LaBianca S, Brikell I, Helenius D, et al. Polygenic profiles define aspects of clinical heterogeneity in attention deficit hyperactivity disorder. *Nat Genet*. 2024;56(2):234-244. doi:10.1038/s41588-023-01593-7

32. Bybjerg-Grauholm J, Pedersen CB, Bækvad-Hansen M, et al. The iPSYCH2015 case-cohort sample: updated directions for unravelling genetic and environmental architectures of severe mental disorders. *medRxiv*. Preprint published online December 2, 2020. doi:10.1101/2020.11.30.20237768

33. Pedersen CB, Bybjerg-Grauholm J, Pedersen MG, et al. The iPSYCH2012 case-cohort sample: new directions for unravelling genetic and environmental architectures of severe mental disorders. *Mol Psychiatry*. 2018;23(1):6-14. doi:10.1038/mp.2017.196

34. Mors O, Perto GP, Mortensen PB. The Danish psychiatric central research register. *Scand J Public Health*. 2011;39(7)(suppl):54-57. doi:10.1177/1403494810395825

35. Nørgaard-Pedersen B, Hougaard DM. Storage policies and use of the Danish Newborn Screening Biobank. *J Inherit Metab Dis*. 2007;30(4):530-536. doi:10.1007/s10545-007-0631-x

36. Lam M, Awasthi S, Watson HJ, et al. RICOPILI: Rapid imputation for consortias pipeline. *Bioinformatics*. 2020;36(3):930-933. doi:10.1093/bioinformatics/btz633

37. McCarthy S, Das S, Kretzschmar W, et al; Haplotype Reference Consortium. A reference panel of 64,976 haplotypes for genotype imputation. *Nat Genet*. 2016;48(10):1279-1283. doi:10.1038/ng.3643

38. Demontis D, Walters RK, Martin J, et al; ADHD Working Group of the Psychiatric Genomics Consortium (PGC); Early Lifecourse & Genetic Epidemiology (EAGLE) Consortium; 23andMe Research Team. Discovery of the first genome-wide significant risk loci for attention deficit/hyperactivity disorder. *Nat Genet*. 2019;51(1):63-75. doi:10.1038/s41588-018-0269-7

39. Grove J, Ripke S, Als TD, et al; Autism Spectrum Disorder Working Group of the Psychiatric Genomics Consortium; BUPGEN; Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium; 23andMe Research Team. Identification of common genetic risk variants for autism spectrum disorder. *Nat Genet*. 2019;51(3):431-444. doi:10.1038/s41588-019-0344-8

40. Stahl EA, Breen G, Forstner AJ, et al; eQTLGen Consortium; BIOS Consortium; Bipolar Disorder Working Group of the Psychiatric Genomics Consortium. Genome-wide association study identifies 30 loci associated with bipolar disorder. *Nat Genet*. 2019;51(5):793-803. doi:10.1038/s41588-019-0397-8

41. Howard DM, Adams MJ, Clarke TK, et al; 23andMe Research Team; Major Depressive Disorder Working Group of the Psychiatric Genomics Consortium. Genome-wide meta-analysis of depression identifies 102 independent variants and highlights the importance of the prefrontal brain regions. *Nat Neurosci*. 2019;22(3):343-352. doi:10.1038/s41593-018-0326-7

42. Ripke S, Neale BM, Corvin A, et al; Schizophrenia Working Group of the Psychiatric Genomics Consortium. Biological insights from 108 schizophrenia-associated genetic loci. *Nature*. 2014;511(7510):421-427. doi:10.1038/nature13595

43. Okbay A, Wu Y, Wang N, et al; 23andMe Research Team; Social Science Genetic Association Consortium. Polygenic prediction of educational attainment within and between families from genome-wide association analyses in 3 million individuals. *Nat Genet*. 2022;54(4):437-449. doi:10.1038/s41588-022-01016-z

44. Savage JE, Jansen PR, Stringer S, et al. Genome-wide association meta-analysis in 269,867 individuals identifies new genetic and functional links to intelligence. *Nat Genet*. 2018;50(7):912-919. doi:10.1038/s41588-018-0152-6

45. Karlsson Linnér R, Biroli P, Kong E, et al; 23and Me Research Team; eQTLgen Consortium; International Cannabis Consortium; Social Science Genetic Association Consortium. Genome-wide association analyses of risk tolerance and risky behaviors in over 1 million individuals identify hundreds of loci and shared genetic influences. *Nat Genet*. 2019;51(2):245-257. doi:10.1038/s41588-018-0309-3

46. Nagel M, Jansen PR, Stringer S, et al; 23andMe Research Team. Meta-analysis of genome-wide association studies for neuroticism in 449,484 individuals identifies novel genetic loci and pathways. *Nat Genet*. 2018;50(7):920-927. doi:10.1038/s41588-018-0151-7

47. Hatoum AS, Johnson EC, Colbert SMC, et al. The addiction risk factor: a unitary genetic vulnerability characterizes substance use disorders and their associations with common correlates. *Neuropsychopharmacology*. 2022;47(10):1739-1745. doi:10.1038/s41386-021-01209-w

48. Albiñana C, Grove J, McGrath JJ, et al. Leveraging both individual-level genetic data and GWAS summary statistics increases polygenic prediction. *Am J Hum Genet*. 2021;108(6):1001-1011. doi:10.1016/j.ajhg.2021.04.014

49. Privé F, Arbel J, Vilhjálmsson BJ. LDpred2: better, faster, stronger. *Bioinformatics*. 2021;36(22-23):5424-5431. doi:10.1093/bioinformatics/btaa1029

50. Clayton D, Schifflers E. Models for temporal variation in cancer rates: I: age-period and age-cohort models. *Stat Med*. 1987;6(4):449-467. doi:10.1002/sim.4780060405

© 2026 American Medical Association. All rights reserved, including those for text and data mining, AI training, and similar technologies.

Downloaded from jamanetwork.com by SUNY Stony Brook user on 10/21/2026